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T4 RAM Disk

Import Thư viện

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!pip install diffusers accelerate safetensors transformers

```
Requirement already satisfied: diffusers in /usr/local/lib/python3.11/dist-packages (0.32.2)
Requirement already satisfied: accelerate in /usr/local/lib/python3.11/dist-packages (1.3.0)
Requirement already satisfied: safetensors in /usr/local/lib/python3.11/dist-packages (0.5.3)
Requirement already satisfied: transformers in /usr/local/lib/python3.11/dist-packages (4.48.3)
Requirement already satisfied: importlib-metadata in /usr/local/lib/python3.11/dist-packages (from diffusers) (8.6.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from diffusers) (3.17.0)
Requirement already satisfied: huggingface-hub<0.23.2 in /usr/local/lib/python3.11/dist-packages (from diffusers) (0.28.1)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from diffusers) (1.26.4)
Requirement already satisfied: regex<2019.12.17 in /usr/local/lib/python3.11/dist-packages (from diffusers) (2024.11.6)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from diffusers) (2.32.3)
Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packages (from diffusers) (11.1.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from accelerate) (24.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.11/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.11/dist-packages (from accelerate) (6.0.2)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from accelerate) (2.5.1+cu124)
Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.21.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.11/dist-packages (from transformers) (4.67.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.23.2->diffusers) (2024.10.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.23.2->diffusers) (4.12.2)
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (3.4.2)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (3.1.5)
Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch>=2.0.0->accelerate)
  Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cuda-runtime-cu12==12.4.127 (from torch>=2.0.0->accelerate)
  Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
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  Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch>=2.0.0->accelerate)
  Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cublas-cu12==12.4.5.8 (from torch>=2.0.0->accelerate)
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Collecting nvidia-cufft-cu12==11.2.1.3 (from torch>=2.0.0->accelerate)
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Collecting nvidia-curand-cu12==10.3.5.147 (from torch>=2.0.0->accelerate)
  Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch>=2.0.0->accelerate)
  Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cuspars-cu12==12.3.1.170 (from torch>=2.0.0->accelerate)
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Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (2.21.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (12.4.127)
Collecting nvidia-nvjitlink-cu12==12.4.127 (from torch>=2.0.0->accelerate)
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Requirement already satisfied: triton==3.1.0 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (3.1.0)
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages (from torch>=2.0.0->accelerate) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy==1.13.1->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.11/dist-packages (from importlib-metadata->diffusers) (3.21.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->diffusers) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests->diffusers) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests->diffusers) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests->diffusers) (2025.1.31)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from Jinja2->torch>=2.0.0->accelerate) (3.0.2)
Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl (363.4 MB)
363.4/363.4 MB 4.3 MB/s eta 0:00:00
Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (13.8 MB)
13.8/13.8 MB 108.2 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (24.6 MB)
```

Connect Drive

[2] from google.colab import drive
drive.mount('/content/drive')

Mounted at /content/drive

Test 1

```
[3] # Thư viện cần thiết
import PIL
import torch
from PIL import ImageOps
from diffusers import StableDiffusionInstructPix2PixPipeline, EulerAncestralDiscreteScheduler
import matplotlib.pyplot as plt

# Load mô hình Instruct-Pix2Pix
model_id = "timbrooks/instruct-pix2pix"
pipe = StableDiffusionInstructPix2PixPipeline.from_pretrained(model_id, torch_dtype=torch.float16, safety_checker=None)
pipe.to("cuda")
pipe.scheduler = EulerAncestralDiscreteScheduler.from_config(pipe.scheduler.config)

# Đọc ảnh từ Google Drive
input_image_path = '/content/drive/MyDrive/OJT/images/IMG_1683.png' #thay path
image = PIL.Image.open(input_image_path)

# Định dạng RGB
image = ImageOps.exif_transpose(image)
image = image.convert("RGB")

# Câu lệnh prompt
prompt = "photo white balance"

# Tiến hành tạo ảnh output với mô hình
images = pipe(prompt, image=image, num_inference_steps=10, image_guidance_scale=1).images
```

```
# Lưu ảnh output
output_image_path = '/content/drive/MyDrive/OJT/images/result_colab/output_1683.jpg' # Thay đổi path
images[0].save(output_image_path) # Lưu ảnh output
# Hiển thị ảnh input và output
fig, axes = plt.subplots(1, 2, figsize=(12, 6))

# Hiển thị ảnh input bên trái
axes[0].imshow(image)
axes[0].set_title('Input Image')
axes[0].axis('off') # Ẩn trục

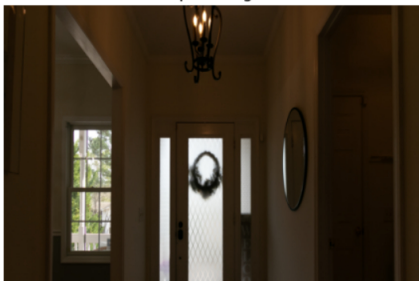
# Hiển thị ảnh output bên phải
axes[1].imshow(images[0])
axes[1].set_title('Output Image')
axes[1].axis('off') # Ẩn trục

# Hiển thị kết quả
plt.show()
```

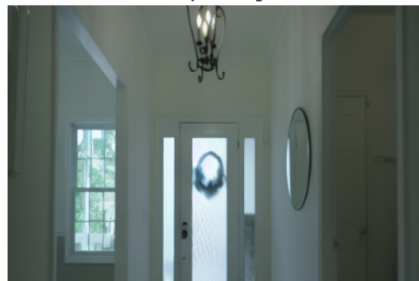
The cache for model files in Transformers v4.22.0 has been updated. Migrating your old cache. This is a one-time only operation. You can interrupt this and resume the migration later

```
0/0 [00:00<?, ?H/s]
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret 'HF_TOKEN' does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
model_index.json: 100% ██████████ 616/616 [00:00<00:00, 37.7kB/s]
Fetching 13 files: 100% ██████████ 13/13 [00:49<00:00, 4.98s/it]
model.safetensors: 100% ██████████ 492M/492M [00:10<00:00, 58.0MB/s]
scheduler%2Fscheduler_config.json: 100% ██████████ 569/569 [00:00<00:00, 11.2kB/s]
(...)ure_extractor%2Fpreprocessor_config.json: 100% ██████████ 518/518 [00:00<00:00, 8.17kB/s]
tokenizer%2Fspecial_tokens_map.json: 100% ██████████ 472/472 [00:00<00:00, 6.12kB/s]
tokenizer%2Fmerges.txt: 100% ██████████ 525k/525k [00:00<00:00, 5.72MB/s]
tokenizer%2Fvocab.json: 100% ██████████ 1.06M/1.06M [00:01<00:00, 543kB/s]
text_encoder%2Fconfig.json: 100% ██████████ 617/617 [00:00<00:00, 7.90kB/s]
tokenizer%2Ftokenizer_config.json: 100% ██████████ 806/806 [00:00<00:00, 8.19kB/s]
diffusion_pytorch_model.safetensors: 100% ██████████ 3.44G/3.44G [00:48<00:00, 88.6MB/s]
diffusion_pytorch_model.safetensors: 100% ██████████ 335M/335M [00:08<00:00, 73.3MB/s]
vae%2Fconfig.json: 100% ██████████ 553/553 [00:00<00:00, 32.0kB/s]
unet%2Fconfig.json: 100% ██████████ 1.02k/1.02k [00:00<00:00, 6.86kB/s]
Loading pipeline components...: 100% ██████████ 6/6 [00:23<00:00, 3.08s/it]
100% ██████████ 10/10 [00:02<00:00, 4.79it/s]
```

Input Image



Output Image



Test 2:

```
[4] # Thư viện cần thiết
import PIL
import torch
from PIL import ImageOps
from diffusers import StableDiffusionInstructPix2PixPipeline, EulerAncestralDiscreteScheduler
import matplotlib.pyplot as plt

# Load mô hình Instruct-Pix2Pix
model_id = "timbrooks/instruct-pix2pix"
pipe = StableDiffusionInstructPix2PixPipeline.from_pretrained(model_id, torch_dtype=torch.float16, safety_checker=None)
pipe.to("cuda")
pipe.scheduler = EulerAncestralDiscreteScheduler.from_config(pipe.scheduler.config)

# Đọc ảnh từ Google Drive
input_image_path = '/content/drive/MyDrive/OJT/images/IMG_1684.png' #thay path
image = PIL.Image.open(input_image_path)

# Định dạng RGB
image = ImageOps.exif_transpose(image)
image = image.convert("RGB")

# Câu lệnh prompt
prompt = "photo white balance"

# Tiến hành tạo ảnh output với mô hình
images = pipe(prompt, image=image, num_inference_steps=10, image_guidance_scale=1).images

# Lưu ảnh output vào Google Drive
```

```

output_image_path = '/content/drive/MyDrive/OJT/images/result_colab/output_1684.jpg' # Thay đổi path
images[0].save(output_image_path) # Lưu ảnh output
# Hiển thị ảnh input và output
fig, axes = plt.subplots(1, 2, figsize=(12, 6))

# Hiển thị ảnh input bên trái
axes[0].imshow(image)
axes[0].set_title('Input Image')
axes[0].axis('off') # Ẩn trục

# Hiển thị ảnh output bên phải
axes[1].imshow(images[0])
axes[1].set_title('Output Image')
axes[1].axis('off') # Ẩn trục

# Hiển thị kết quả
plt.show()

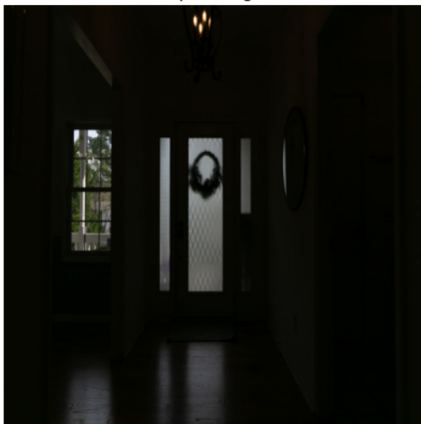
```



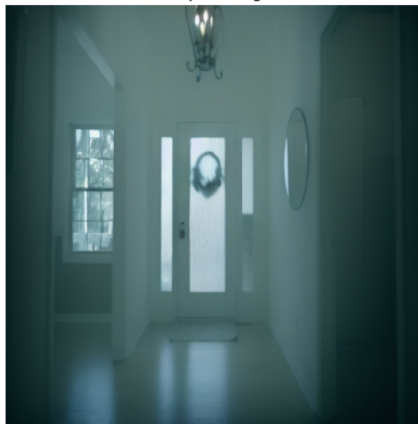
Loading pipeline components...: 100% 6/6 [00:07<00:00, 1.22s/it]

100% 10/10 [00:02<00:00, 4.63it/s]

Input Image



Output Image



Test 3

```

[5] # Thư viện cần thiết
import PIL
import torch
from PIL import ImageOps
from diffusers import StableDiffusionInstructPix2PixPipeline, EulerAncestralDiscreteScheduler
import matplotlib.pyplot as plt

# Load mô hình Instruct-Pix2Pix
model_id = "timbrooks/instruct-pix2pix"
pipe = StableDiffusionInstructPix2PixPipeline.from_pretrained(model_id, torch_dtype=torch.float16, safety_checker=None)
pipe.to("cuda")
pipe.scheduler = EulerAncestralDiscreteScheduler.from_config(pipe.scheduler.config)

# Đọc ảnh từ Google Drive
input_image_path = '/content/drive/MyDrive/OJT/images/IMG_1685.png' #thay path
image = PIL.Image.open(input_image_path)

# Định dạng RGB
image = ImageOps.exif_transpose(image)
image = image.convert("RGB")

# Câu lệnh prompt
prompt = "photo white balance"

# Tiến hành tạo ảnh output với mô hình
images = pipe(prompt, image=image, num_inference_steps=10, image_guidance_scale=1).images

# Lưu ảnh output vào Google Drive
output_image_path = '/content/drive/MyDrive/OJT/images/result_colab/output_1685.jpg' # Thay đổi path
images[0].save(output_image_path) # Lưu ảnh output
# Hiển thị ảnh input và output
fig, axes = plt.subplots(1, 2, figsize=(12, 6))

# Hiển thị ảnh input bên trái
axes[0].imshow(image)
axes[0].set_title('Input Image')
axes[0].axis('off') # Ẩn trục

# Hiển thị ảnh output bên phải
axes[1].imshow(images[0])
axes[1].set_title('Output Image')
axes[1].axis('off') # Ẩn trục

# Hiển thị kết quả
plt.show()

```



Loading pipeline components...: 100% 6/6 [00:04<00:00, 1.48it/s]

100% 10/10 [00:02<00:00, 4.47it/s]

Input Image



Output Image





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✓ 9s completed at 3:33 PM

